$Z\phi$-classes for $\mathrm{PerfectGroup}(1920, 3)$: continuation $64 \le n \le 127$

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$Z(P) \cong C_2$, $\mathrm{Out}(P) \cong \mathrm{Sym}(4)$

n	Runtime (s)	$\#Z\phi$	Cumulative
64	2734.748	735,590	772,803
65	0.162	1	772,804
66	2.150	32	772,836
67	0.019	1	772,837
68	1.870	104	772,941
69	0.041	2	772,943
70	1.672	24	772,967
71	0.019	1	772,968
72	72.985	3,450	776,418
73	0.016	1	776,419
74	0.207	12	776,431
75	0.317	6	776,437
76	1.325	96	776,533
77	0.032	1	776,534
78	1.607	48	776,582
79	0.015	1	776,583
80	86.884	9,708	786,291
81	4.882	46	786,337
82	0.203	12	786,349
83	0.016	1	786,350
84	11.398	468	786,818
85	0.037	1	786,819

n	Runtime (s)	$\#Z\phi$	Cumulative
86	0.248	12	786,831
87	0.046	2	786,833
88	8.968	742	787,575
89	0.024	1	787,576
90	4.478	84	787,660
91	0.047	1	787,661
92	1.278	96	787,757
93	0.201	4	787,761
94	0.253	12	787,773
95	0.037	1	787,774
96	1311.322	191,824	979,598
97	0.017	1	979,599
98	1.134	30	979,629
99	0.182	4	979,633
100	10.208	348	979,981
101	0.016	1	979,982
102	1.173	32	980,014
103	0.020	1	980,015
104	13.144	830	980,845
105	0.758	4	980,849
106	0.250	12	980,861

n	Runtime (s)	$\#Z\phi$	Cumulative
107	0.017	1	980,862
108	60.148	1,424	982,286
109	0.016	1	982,287
110	1.513	36	982,323
111	0.142	4	982,327
112	81.758	8,888	991,215
113	0.017	1	991,216
114	1.677	48	991,264
115	0.034	1	991,265
116	1.785	104	991,369
117	0.611	9	991,378
118	0.242	12	991,390
119	0.039	1	991,391
120	74.490	3,917	995,308
121	0.084	2	995,310
122	0.280	12	995,322
123	0.042	2	995,324
124	1.512	96	995,420
125	0.474	5	995,425
126	8.061	140	995,565
127	0.019	1	995,566

Summary. This continuation covers $64 \le n \le 127$. It contains **958,353** $Z\phi$-classes and required **4507.370 s** (about 75 min 7.4 s) of measured GAP runtime. The cumulative column continues the preceding table: it starts from 37,213 classes at $n = 63$, reaches **995,566** classes at $n = 127$, and therefore gives the cumulative count for the complete range $2 \le n \le 127$. Combining both tables, the measured runtime is **4814.839 s** (about 80 min 14.8 s). Each row counts the $Z\phi$-classes attached to auxiliary groups S of order $2n$; runtimes are empirical GAP timings for the validated v7 implementation.

Runtime concentration for $\mathrm{PerfectGroup}(1920, 3)$**,** $64 \le n \le 127$

Claude Archer

$Z(P) \cong C_2, \quad \mathrm{Out}(P) \cong \mathrm{Sym}(4)$

n	Runtime (s)	$\#Z\phi$-classes	Share of range runtime	Cumulative share
64	2734.748	735,590	60.67%	60.67%
96	1311.322	191,824	29.09%	89.77%
80	86.884	9,708	1.93%	91.69%
112	81.758	8,888	1.81%	93.51%
120	74.490	3,917	1.65%	95.16%
72	72.985	3,450	1.62%	96.78%
108	60.148	1,424	1.33%	98.11%
104	13.144	830	0.29%	98.41%
84	11.398	468	0.25%	98.66%
100	10.208	348	0.23%	98.88%
88	8.968	742	0.20%	99.08%
126	8.061	140	0.18%	99.26%

Reading the timing profile. The computation is even more concentrated than in the preceding range. The two cases $n = 64$ and $n = 96$ alone account for **89.77%** of the runtime for $64 \le n \le 127$. The six slowest values $(64, 96, 80, 112, 120, 72)$ account for **96.78%**. Thus the large majority of values in this interval are computationally inexpensive.

The two dominant cases. For $n = 64$, the auxiliary order is $|S| = 128$: all 2,328 groups of order 128 are scanned, giving **735,590** classes in 2,734.748 s. The audit records 289,320 normal subgroups, 555,820 candidate pairs and 244,341 pair orbits; `NormalSubgroups(S)` alone accounts for 44.22% of the runtime and the generator action on admissible subgroups for 23.99%. For $n = 96$, $|S| = 192$ and 1,543 groups are scanned, giving **191,824** classes in 1,311.322 s. Its audit records 135,814 normal subgroups, 162,735 candidate pairs and 71,963 pair orbits. Here the main costs are again `NormalSubgroups(S)` (495.535 s) and `AutomorphismGroup(S)` (318.958 s).

Algorithmic interpretation. The v7 orbit/stabilizer machinery is no longer the principal bottleneck. The expensive cases are driven mainly by the construction of normal-subgroup lattices and by the induced action of automorphism generators. The elementary-abelian fast path already removes the extreme case C_2^r. At $n = 64$, the family near the top of the order-128 library, including $C_4 \times C_2^5$, is expensive but represents only a small fraction of the total runtime. A future optimization should therefore target the direct construction of admissible kernels $K \lhd S$ with quotient S/K isomorphic to a subgroup of S_4, rather than add a fast path only for $C_4 \times C_2^r$.

Scope and reproducibility. Together with the preceding table for $2 \le n \le 63$, these results give the complete validated v7 count through $n = 127$: **995,566** $Z\phi$**-classes**. The raw TSV/checkpoint audit files should be retained with the GAP source so that the published table can be regenerated and the timing profile independently inspected.